B Quasi-Degenerate Perturbation Theory

Quasi-degenerate perturbation theory ("Löwdin partitioning", [1, 2, 3])[1] is a general and powerful method for the approximate diagonalization of time-independent Hamiltonians H. It is particularly suited for the perturbative diagonalization of $\boldsymbol{k} \cdot \boldsymbol{p}$ multiband Hamiltonians, but it can also be used for many other problems in quantum mechanics. Quasi-degenerate perturbation theory is closely related to conventional stationary perturbation theory. However, it is more powerful because we need not distinguish between nondegenerate and degenerate perturbation theory. As this method is not well known in quantum mechanics, we include a more detailed description of it here.

The Hamiltonian H is expressed as the sum of two parts: a Hamiltonian H^0 with known eigenvalues E_n and eigenfunctions $|\psi_n\rangle$, and H', which is treated as a perturbation:

$$H = H^0 + H' \,. \tag{B.1}$$

We assume that we can divide the set of eigenfunctions $\{|\psi_n\rangle\}$ into weakly interacting subsets A and B such that we are interested only in the set A and not in B. Quasi-degenerate perturbation theory is based on the idea that we construct a unitary operator e^{-S} such that for the transformed Hamiltonian

$$\tilde{H} = \mathrm{e}^{-S} H \mathrm{e}^{S} \,, \tag{B.2}$$

the matrix elements $\langle\psi_m|\tilde{H}|\psi_l\rangle$ between states $|\psi_m\rangle$ from set A and states $|\psi_l\rangle$ from set B vanish up to the desired order of H'. We can depict the removal of the off-diagonal elements of H as shown in Fig. B.1.

We begin by writing H as

$$H = H^0 + H' = H^0 + H^1 + H^2 \,, \tag{B.3}$$

where H^1 has nonzero matrix elements only between the eigenfunctions $|\psi_n\rangle$ *within* the sets A and B, whereas H^2 has nonzero matrix elements only *between* the sets A and B as shown in Fig. B.2. Obviously, we must construct

[1] Quasi-degenerate perturbation theory has been applied to various physical problems. It was used by Van Vleck to study the spectra of diatomic molecules [4]. Furthermore, it is essentially equivalent to the Foldy–Wouthuysen transformation [5] used in the context of relativistic quantum mechanics and to the Schrieffer–Wolff transformation [6] used in the context of the Anderson and Kondo models. A general discussion of unitary transformations is given in [7].

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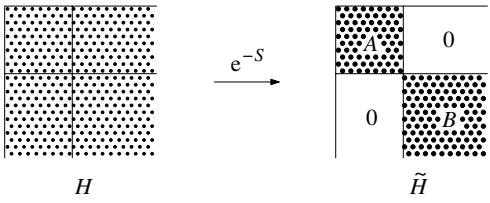


Fig. B.1. Removal of off-diagonal elements of H

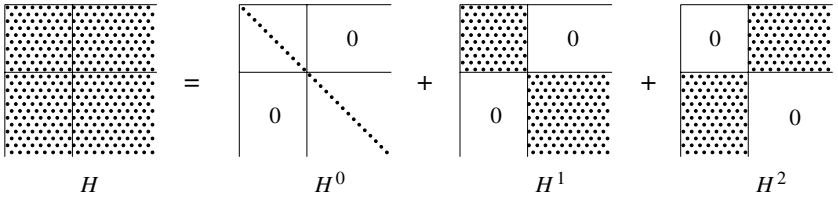


Fig. B.2. Representation of H as $H^0 + H^1 + H^2$

S such that the transformation (B.2) converts H^2 into a block-diagonal form similar to H^1 while keeping the desired block-diagonal form of $H^0 + H^1$. In order to determine the operator S, we expand e^S in a series

$$\mathrm{e}^S = 1 + S + \frac{1}{2!}S^2 + \frac{1}{3!}S^3 \ldots \tag{B.4}$$

and construct S by successive approximations. Substituting (B.4) into (B.2), and noting that the operator S must be anti-Hermitian, i.e. $S^\dagger = -S$, we obtain

$$\tilde{H} = \sum_{j=0}^{\infty} \frac{1}{j!} \big[H, S\big]^{(j)} = \sum_{j=0}^{\infty} \frac{1}{j!} \big[H^0 + H^1, S\big]^{(j)} + \sum_{j=0}^{\infty} \frac{1}{j!} \big[H^2, S\big]^{(j)} \,, \tag{B.5}$$

where the commutators $\big[A, B\big]^{(j)}$ are defined by

$$\big[A, B\big]^{(j)} = [\ldots[[\underbrace{A, B], B], \ldots B}_{j \text{ times}}] \,. \tag{B.6}$$

Since S must be non-block-diagonal like H^2, the block-diagonal part $\tilde{H}_\mathrm{d}$ of $\tilde{H}$ contains the terms $\big[H^0 + H^1, S\big]^{(j)}$ with even j and $\big[H^2, S\big]^{(j)}$ with odd j:

$$\tilde{H}_\mathrm{d} = \sum_{j=0}^{\infty} \frac{1}{(2j)!} \big[H^0 + H^1, S\big]^{(2j)} + \sum_{j=0}^{\infty} \frac{1}{(2j+1)!} \big[H^2, S\big]^{(2j+1)} \,. \tag{B.7}$$

Conversely, the non-block-diagonal part $\tilde{H}_\mathrm{n}$ of $\tilde{H}$ must contain the terms $\big[H^0 + H^1, S\big]^{(j)}$ with odd j and $\big[H^2, S\big]^{(j)}$ with even j:

$$\tilde{H}_\mathrm{n} = \sum_{j=0}^{\infty} \frac{1}{(2j+1)!} \big[H^0 + H^1, S\big]^{(2j+1)} + \sum_{j=0}^{\infty} \frac{1}{(2j)!} \big[H^2, S\big]^{(2j)} \,. \tag{B.8}$$

Now S is defined by the condition that the non-block-diagonal part $\tilde{H}_{\mathrm{n}}$ of $\tilde{H}$ vanishes:

$$\tilde{H}_{\mathrm{n}} = 0 \,. \tag{B.9}$$

Choosing the ansatz

$$S = S^{(1)} + S^{(2)} + S^{(3)} + \dots \,, \tag{B.10}$$

we can use (B.8) and (B.9) to derive the following equations for the successive approximations $S^{(j)}$ to S:

$$[H^0, S^{(1)}] = -H^2 \,, \tag{B.11a}$$
$$[H^0, S^{(2)}] = -[H^1, S^{(1)}] \,, \tag{B.11b}$$
$$[H^0, S^{(3)}] = -[H^1, S^{(2)}] - \tfrac{1}{3}[[H^2, S^{(1)}], S^{(1)}] \,, \tag{B.11c}$$
$$\dots \quad = \dots \,.$$

Solving (B.11) for $S^{(j)}$ we obtain

$$S^{(1)}_{ml} = -\frac{H'_{ml}}{E_m - E_l} \,, \tag{B.12a}$$

$$S^{(2)}_{ml} = \frac{1}{E_m - E_l}\left[\sum_{m'} \frac{H_{mm'}H_{m'l}}{E_{m'} - E_l} - \sum_{l'} \frac{H_{ml'}H_{l'l}}{E_m - E_{l'}}\right] \,, \tag{B.12b}$$

$$\begin{aligned} S^{(3)}_{ml} = {} & \frac{1}{E_m - E_l} \\ & \times \Bigg[- \sum_{m',\,m''} \frac{H_{mm''}H_{m''m'}H_{m'l}}{(E_{m''} - E_l)(E_{m'} - E_l)} - \sum_{l',\,l''} \frac{H_{ml'}H_{l'l''}H_{l''l}}{(E_m - E_{l''})(E_m - E_{l'})} \\ & + \sum_{l',\,m'} \frac{H_{mm'}H_{m'l'}H_{l'l}}{(E_{m'} - E_l)(E_{m'} - E_{l'})} + \sum_{l',\,m'} \frac{H_{mm'}H_{m'l'}H_{l'l}}{(E_m - E_{l'})(E_{m'} - E_{l'})} \\ & + \frac{1}{3}\sum_{l',\,m'} \frac{H_{ml'}H_{l'm'}H_{m'l}}{(E_{m'} - E_{l'})(E_{m'} - E_l)} + \frac{1}{3}\sum_{l',\,m'} \frac{H_{ml'}H_{l'm'}H_{m'l}}{(E_m - E_{l'})(E_{m'} - E_{l'})} \\ & + \frac{2}{3}\sum_{l',\,m'} \frac{H_{ml'}H_{l'm'}H_{m'l}}{(E_m - E_{l'})(E_{m'} - E_l)} \Bigg] \,, \end{aligned} \tag{B.12c}$$

$$\dots \quad = \dots \,.$$

Here the indices m, m', m'' correspond to states in set A, the indices l, l', l'' correspond to states in set B, and

$$H_{ml} \equiv \langle \psi_m | H | \psi_l \rangle \,. \tag{B.13}$$

Inserting (B.12) into (B.7), we finally obtain the desired equations for the successive approximations to $\tilde{H}$

$$\tilde{H} = H^{(0)} + H^{(1)} + H^{(2)} + H^{(3)} + H^{(4)} + \dots \,, \tag{B.14}$$

where $H^{(j)}$ up to $j = 4$ are given in (B.15). We would like to emphasize that all the energy denominators in (B.15) refer to pairs of states with one state from set A and the other state from set B. The sets A and B will always be choosen such that they are separated in energy. Therefore, unlike the situation for conventional stationary perturbation theory, we can apply (B.15) to systems where the states in set A may contain arbitrary (possibly unknown) exact or approximate degeneracies.[2] It is thus particularly suited to computer algebra systems such as Maple[3] and Mathematica.[4]

It is important to note that the derivation of the final equation (B.15) remains valid when the Hamiltonian H contains a matrix of operators $H'_{nn'}$ plus some diagonal energies $E_n \delta_{nn'}$

$$H_{nn'} = E_n \delta_{nn'} + H'_{nn'} \,. \tag{B.16}$$

In this case, S is also a matrix of operators the elements of which are still defined by (B.12). Likewise, the unitary transformation (B.2) can still be defined via (B.5). However, it becomes clear from (B.5) and (B.11) that when $H'_{nn'}$ are operators we must take into account the noncommutativity of different matrix elements $H'_{nn'}$. Therefore, in (B.12) and (B.15), we have to evaluate the products of matrix elements $H'_{nn'}$ *in the given order*.

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[2] In [8], Sakurai presents a three-level system (problem 5.12, p. 348), which he calls a "tricky problem, because the degeneracy between the first and second state is not removed in first order." Using Löwdin partitioning, this problem can readily be solved up to any order.

[3] Maple is a registered trademark of Waterloo Maple Inc.

[4] Mathematica is a registered trademark of Wolfram Research Inc.

$$H^{(0)}_{mm'} = H^0_{mm'} \ , \tag{B.15a}$$

$$H^{(1)}_{mm'} = H'_{mm'} \ , \tag{B.15b}$$

$$H^{(2)}_{mm'} = \frac{1}{2}\sum_l H'_{ml}H'_{lm'}\left[\frac{1}{E_m - E_l} + \frac{1}{E_{m'} - E_l}\right] \ , \tag{B.15c}$$

$$\begin{aligned} H^{(3)}_{mm'} = & -\frac{1}{2}\sum_{l,\,m''}\left[\frac{H'_{ml}H'_{lm''}H'_{m''m'}}{(E_{m'} - E_l)(E_{m''} - E_l)} + \frac{H'_{mm''}H'_{m''l}H'_{lm'}}{(E_m - E_l)(E_{m''} - E_l)}\right] \\ & + \frac{1}{2}\sum_{l,\,l'} H'_{ml}H'_{ll'}H'_{l'm'}\left[\frac{1}{(E_m - E_l)(E_m - E_{l'})} + \frac{1}{(E_{m'} - E_l)(E_{m'} - E_{l'})}\right] \ , \end{aligned} \tag{B.15d}$$

$$\begin{aligned} H^{(4)}_{mm'} = & \quad \frac{1}{2}\sum_{l,\,m'',\,m'''} \frac{1}{(E_{m''} - E_l)(E_{m'''} - E_l)}\left[\frac{H'_{mm''}H'_{m''m'''}H'_{m'''l}H'_{lm'}}{E_m - E_l} + \frac{H'_{ml}H'_{lm''}H'_{m''m'''}H'_{m'''m'}}{E_{m'} - E_l}\right] \\ & - \frac{1}{2}\sum_{l,\,l',\,m''}\left[\frac{H'_{ml}H'_{ll'}H'_{l'm''}H'_{m''m'}}{(E_{m'} - E_l)(E_{m''} - E_{l'})}\left(\frac{1}{E_{m''} - E_l} + \frac{1}{E_{m'} - E_{l'}}\right) + \frac{H'_{mm''}H'_{m''l}H'_{ll'}H'_{l'm'}}{(E_m - E_{l'})(E_{m''} - E_l)}\left(\frac{1}{E_{m''} - E_{l'}} + \frac{1}{E_m - E_l}\right)\right] \\ & - \frac{1}{24}\sum_{l,\,l',\,m''} H'_{ml}H'_{lm''}H'_{m''l'}H'_{l'm'}\left[\frac{8}{(E_m - E_l)(E_m - E_{l'})(E_{m''} - E_{l'})} + \frac{8}{(E_{m'} - E_l)(E_{m'} - E_{l'})(E_{m''} - E_l)}\right. \\ & \qquad + \frac{4}{(E_m - E_{l'})(E_{m''} - E_l)}\left(\frac{1}{E_m - E_l} + \frac{1}{E_{m''} - E_{l'}}\right) + \frac{4}{(E_{m'} - E_l)(E_{m''} - E_{l'})}\left(\frac{1}{E_{m'} - E_{l'}} + \frac{1}{E_{m''} - E_l}\right) \\ & \qquad \left. - \frac{1}{(E_{m''} - E_l)(E_{m''} - E_{l'})}\left(\frac{1}{E_m - E_l} + \frac{1}{E_{m'} - E_{l'}}\right) - \frac{3}{(E_m - E_l)(E_{m'} - E_{l'})}\left(\frac{1}{E_{m''} - E_l} + \frac{1}{E_{m''} - E_{l'}}\right)\right] \\ & + \frac{1}{2}\sum_{l,\,l',\,l''} H'_{ml}H'_{ll'}H'_{l'l''}H'_{l''m'}\left[\frac{1}{(E_m - E_l)(E_m - E_{l'})(E_m - E_{l''})} + \frac{1}{(E_{m'} - E_l)(E_{m'} - E_{l'})(E_{m'} - E_{l''})}\right] , \end{aligned} \tag{B.15e}$$

$$\ldots \quad = \ldots$$